LabelLens: Seeing a Bluesky Post Through Other People's Moderation

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ABSTRACT

On Bluesky, moderation is *stackable*: anyone can run a labeling service, and each user chooses which of these *labelers* to trust. Two people can therefore see the same post very differently, and nobody, including its author, sees the whole picture. We present LabelLens, a serverless web tool that asks every public labeler about a post or an account at once and renders the item through several subscription sets with the official client's moderation code, side by side. A companion page reports a network-wide measurement of all 571 declared labelers. Only 199 of their 493 published endpoints answer queries, and 65 % of those allow a browser to read the answer. Of the labelled posts in a uniform sample, 42.6 % look different to a user subscribed to the twenty most-liked protective labelers than to a logged-out visitor, 99.7 % of them because of a single service, while the twenty most-liked labelers overall change only 0.9 %. LabelLens runs from a link, with no login and no server.

CCS CONCEPTS

• **Information systems** → **Social networking sites**; • **Human-centered computing** → *Collaborative and social computing systems and tools.*

KEYWORDS

decentralized social media, Bluesky, AT Protocol, content moderation, labeling, transparency

1 INTRODUCTION

Bluesky replaces a single platform moderator with *labels* [3, 8]. A labeler is an account that signs short values such as `spam` or `ai-imagery` and attaches them to posts or accounts [1]. Every client applies Bluesky's own labeler, and a signed-in user may subscribe to up to twenty more and decide, per label, whether to hide, warn or ignore. This realises the long-standing idea of moderation *middleware* [7], but it also fragments what people see. An author cannot tell which labels follow their account, a reader cannot tell what other readers are shown, and researchers have so far studied either the network's infrastructure [2] or Bluesky's own labeler [9], which explicitly leaves third-party labelers to future work.

LabelLens makes this fragmentation visible for one item at a time. Given a post link or a handle, it (i) runs a *census*, asking every labeler in a network-wide directory what it says about the post, its author and the author's profile, and (ii) shows the item through *lenses*, that is, subscription sets evaluated with the moderation code of the official client library. A companion *Network* page summarises a measurement of all labelers. The demonstration offers three things: a tool anyone can run from a link, a census method that copes with a network where most services are unreachable from a browser, and measured evidence of how far the stacks diverge.

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2 LABELLENS

LabelLens is a static site (GitHub Pages) of about 580 lines of JavaScript; every request goes from the visitor's browser to public endpoints. Figure 1 shows the Inspect view.

Directory. Labelers are ordinary accounts that publish an `app.bsky.labeler.service` record and a `#atproto_labeler` endpoint in their DID document [1]. We list the accounts holding such a record through the relay's `listReposByCollection`, resolve each DID document, and ship the resulting directory (493 endpoints) with the site; a GitHub Actions workflow repeats the whole measurement.

Census. For an item, the browser sends one `queryLabels` request per labeler (32 in parallel) for the post, the author's DID (account labels) and the profile record (only the last two for an account). Each answer is folded into the labels in force now: the latest label per (labeler, subject, value) wins, a negation retracts it, and an expired label is dropped. Only labels whose source is the queried labeler are kept (signatures are not verified), because some services replay other labelers' labels. Many labelers answer but send no CORS header, so a page cannot read them. For the 61 services without CORS whose label streams we collected, a snapshot of their labels in force at collection time, sharded into 4,096 static files by a hash of the subject's DID, fills the gap, and those rows are marked as snapshot data. The public site ships the snapshot only for its example accounts, so it republishes no labels about individuals; a local build, as used for a live demo, loads all of it. To show the lenses quickly, the census runs in two phases: first the services that answered at our last measurement (4 s timeout); then those that were offline (4 s) and those that were slow (8 s) are retried, and the lenses are recomputed if any of them answers.

Lenses. A lens is a subscription set. LabelLens passes the census result and the labelers' own label definitions, fetched live, to `moderatePost` or `moderateProfile` from `@atproto/api`, with default preferences and adult content off, and reports the decision for a feed or a profile list: hidden, behind a warning, warning badge, info badge, or shown as is, with the labels that caused it. The four lenses are a *logged-out visitor* (Bluesky's labeler only, which is also all the public read API returns), a *popular stack* (the twenty most-liked labelers), a *protective stack* (the twenty most-liked labelers that declare a label that hides, blurs or warns) and *your stack*, which

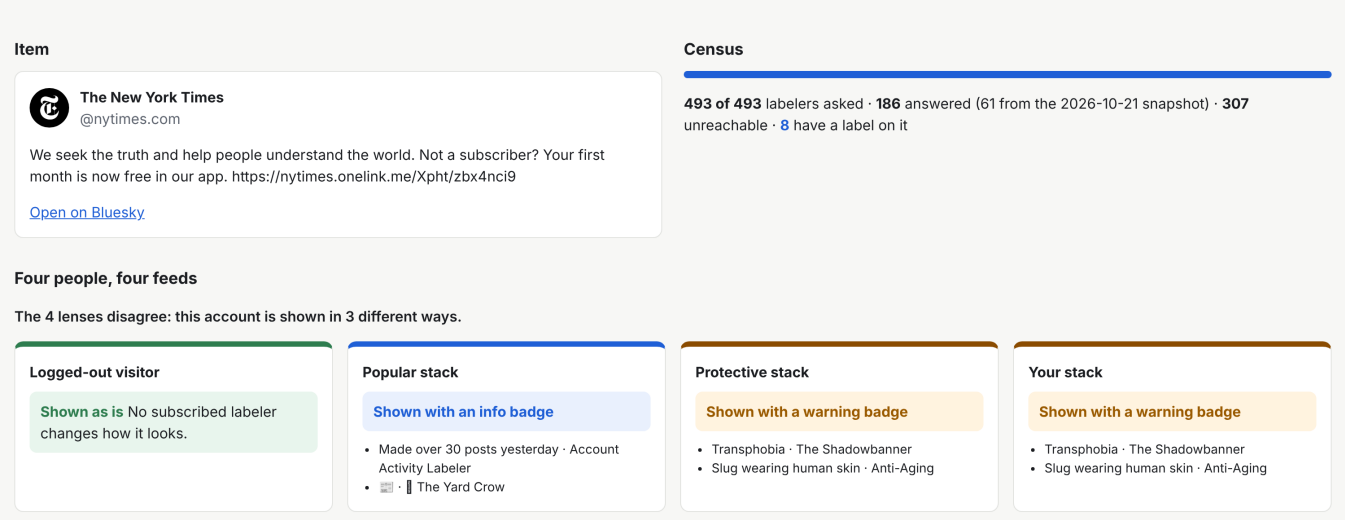


Figure 1: Inspecting a news organisation's account (2026-10-21): 8 labelers label it, and the four lenses show it in 3 different ways. In this run 186 services answered (125 live from the browser, 61 from the snapshot), against 199 endpoints that answered our server-side probe (Section 3). Labels are the labelers' own claims.

starts from the labelers that labelled the item (at most twenty, the client's limit) and can be edited.

Network page. The measurement of Section 3 is published as a sortable table and summary tiles (Figure 2): reachability, label volume, retractions, labels without effect and time to label, per service and in aggregate only.

3 MEASUREMENT AND EVALUATION

Data. A GitHub Actions run on 2026-10-21 listed the labeler directory, probed every endpoint once, and replayed each labeler's full label stream with `subscribeLabels` from sequence zero. For the 6 services whose stream had passed sequence number three million, only the last three million sequence numbers were replayed. An earlier run from a second vantage point (our cloud workspace) gave nearly the same reachability (209 answering, 132 browser-readable), so the failures below are the services', not the vantage point's. The author set the topic and the direction of the experiments; an AI assistant (Anthropic Claude) helped carry them out and analyse them.

Reachability. Of 571 accounts that declare a labeler, 493 publish an endpoint, but only 199 (40 %) answer a `queryLabels` request; the rest cannot be reached (134), fail TLS (32) or answer with an HTTP error or a body that is not JSON (128). Of the answering services, 129 (65 %) send a CORS header. A browser-only client such as a third-party app or LabelLens itself therefore sees at most 26 % of declared endpoints, which is why the census needs its snapshot fallback.

Labels. We collected 59.1 million labels from 158 services (81 % on posts, 17 % on accounts). Volume is concentrated: the five largest services issued 66 % of all labels. 10.2 % of labels carry a value their labeler never declared; clients drop such labels silently, and for 20 services they are the majority. 4.4 % of (subject, value) pairs end retracted. Protective labelers differ widely in speed: among the 24

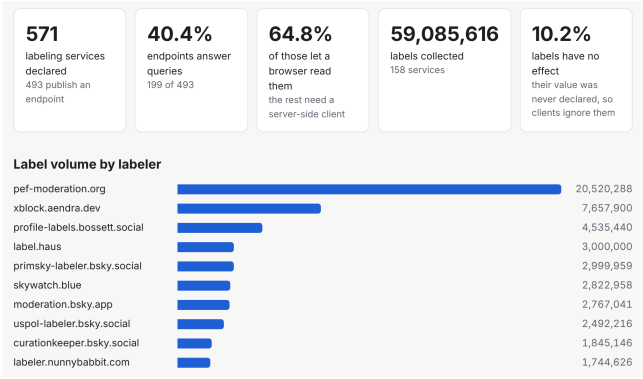


Figure 2: The Network page (2026-10-21): summary tiles and the ten services with the most labels. Services whose stream had passed sequence number three million, such as `moderation.bsky.app`, were replayed from a recent window, so their bars count that window only.

with at least 100 post labels, the median time from post creation to label is under ten seconds for 9 (automated classifiers) and over an hour for 10.

Overlap and divergence. Stacks rarely agree because they rarely overlap: of the 2.9 million accounts labelled by any protective labeler, 3.1 % are labelled by two or more. To measure what this means for readers, we drew a uniform sample of labelled subjects (a subject is kept when its hash is 0 modulo 1,000; 42,454 posts and 4,640 accounts) and evaluated each lens offline with the same moderation code as the site (Table 1). For posts, the protective stack changes the presentation of 42.6 % relative to a logged-out visitor, mostly by adding an info badge, while the popular stack changes only 0.9 %: the most-liked labelers are mostly descriptive (badges,

Table 1: How labelled items are shown under each lens (uniform sample; percent of items).

Lens	Hidden	Warned	Info badge	As is
Posts (n=42,454)				
Logged-out visitor	5.2	0.0	0.0	94.8
Popular stack	5.2	0.0	0.9	93.9
Protective stack	6.6	0.3	40.8	52.2
All that labelled it	10.7	12.7	44.6	32.0
Accounts (n=4,640)				
Logged-out visitor	0.0	0.0	0.0	100.0
Popular stack	0.0	0.5	8.8	90.8
Protective stack	1.1	0.8	9.7	88.5
All that labelled it	3.3	0.9	41.8	54.1

interests, verification), so popularity is a poor guide to protection. The effect is also concentrated: 99.7 % of the changed posts are changed by one service, `pef-moderation.org`, which flags links to other platforms and suspected disinformation sources. Which labelers a user adds matters more than how many, and this is exactly what the census and the editable stack expose for a single item. Subscribing to the labelers that labelled the item (at most twenty) changes 62.8 % of posts.

Census performance. We tried 27 items (posts and accounts from the sample) in headless Chromium from our workspace; 7 sampled posts had since been deleted or hidden, and the other 20 were inspected. The lenses appeared after 3.2 s at the median (4.9 s at most); the recheck of slow services and of those offline at measurement time finished after 15.0 s and added at most 3 answers. At least 118 labelers answered live on every run, and 61 more through the full snapshot.

4 DEMONSTRATION

Visitors open LabelLens at https://seungbum.github.io/labellens/ (no login). A typical walk-through takes three minutes. (1) Paste a post link or a handle, or click an example; the census bar fills as 493 services are asked. (2) Compare the four lens cards: the same item may be shown as is to a logged-out visitor, with an info badge in the popular stack and with a warning in the protective stack, each with the labels that caused it. (3) Open *your stack* and toggle labelers to see which one decides the outcome. (4) Scan the per-labeler list, where struck-through labels were retracted, faded ones expired, and dashed ones have no effect. (5) Switch to the Network page to see how few services are reachable and how unevenly labels are spread. At the conference, attendees can inspect their own accounts on our laptop or their phone; a two-minute video is linked from the site's About page and the repository, https://github.com/seungbum/labellens.

5 RELATED WORK

Bluesky's architecture and composable moderation are described by its developers [3, 8]. Balduf et al. measured the network's components, including a first look at labelers [2]; Singh et al. audited Bluesky's own moderation service [9], and Bono et al. studied blocking as self-moderation [4]. On Mastodon, moderation surfaces as content warnings and instance policies [11]. Transparency work on centralised platforms relies on self-reported statements of reasons in the DSA Transparency Database [5, 10], and the middleware literature argues for third-party moderation services [6, 7]. LabelLens complements these measurement studies with an interactive, per-item view across all third-party labelers, and shows what the middleware model looks like to readers in practice.

6 ETHICAL USE OF DATA

LabelLens reads only public data through public APIs and stores nothing about visitors; queries go from the visitor's browser to Bluesky and the labelers. The measurement replayed public label streams through each service's documented endpoint, one connection per service, with an identifying user agent. We publish aggregate figures per labeling service only: no list of labelled accounts or posts, and the per-labeler table omits label values, which can describe a person (for example pronouns or political affiliation). For services without CORS, the public site ships collected labels only for its organisational example accounts; the full snapshot and the raw label logs stay offline. Labels can be wrong or hostile; showing one does not endorse it, and the interface says so. Examples in the interface and this paper use organisational accounts. No human participants were involved.

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Generative AI disclosure. The author set the topic, directed the design of the system and the measurement, reviewed the results and revised the manuscript, and takes full responsibility for all content and conclusions. Under the author's direction, a generative AI assistant (Anthropic Claude) implemented the system and the measurement scripts, ran the measurement, collected bibliographic metadata and prepared a first draft of the text.

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